Mechanics and Waves

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Mechanics and Waves (A11459)

Short Title: Mechanics and Waves
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author EMOLLOY
Credits 5 **Level:** Introductory

Description of Module / Aims

This module is a calculus-based module. It introduces the differential and integral calculus required for the solution of dynamics problems. Vector calculus and linear algebra are also introduced as required. The physics content is dynamics and waves.

Programmes

PHYI-0009	BSc (Common Entry) (WD_SSCIE_D)	1 / 2 / E
PHYI-0009	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	2 / 3 / M

Indicative Content

- Vectors and scalars: dot and cross products
- Kinematics
- Newton's laws of motion
- Motion under a general potential
- Work and energy
- Uniform circular motion: angular momentum
- Simple harmonic motion including damping: simple treatment of resonance
- Collisions: conservation laws; mechanical energy, total energy and momentum
- Mathematical description of a wave
- Energy, power and intensity in wave motion
- Reflection, refraction, diffraction, interference, polarisation and Doppler effect
- Light waves: geometrical optics; lenses and mirrors
- Sound waves: acoustics
- Practical programme: determination of g; investigation of Newton's second law; friction; measurement of wavelength of light and sound; determination of refractive index; focal lengths of lenses and mirrors

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the fundamentals of mechanics by being able to solve relevant problems.
2. Describe the fundamentals of wave motion by being able to solve relevant problems.
3. Record and analyse experimental data and present results in the form of a report.
4. Communicate results in a presentation to peers.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- Practical/computer labs.

- Additional reading material.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
In-Class Assessment	40%	1,2
Practical	40%	3
Presentation	20%	4

Assessment Criteria

<40%: Student has demonstrated a very limited knowledge of the course material. He/She is unable to carry out critical thinking of subject area.

40%-49%: Student has demonstrated an adequate knowledge of the course material. He/She is able to carry out some critical thinking of subject area.

50%-59%: Shows a reasonable level of understanding of the various principles involved.

60%-69%: Shows good knowledge of the area. Shows ability to reason and analyse problems in this area.

70%-100%: Excellent knowledge of the subject area. Excellent reasoning skills and focused well thought out arguments.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Practical	18	
Independent Learning	87	

Supplementary Material

- Gregory, D.R. _Class Mechanics_. 1st ed. UK: Cambridge University Press, 2006.
- Woolfson, D. _Resonance Applications in Physical Science_. 1st ed. USA: Cambridge University Press, 2015.

Requested Resources

- SCIENCE LAB: Physics